

NUTRIENTS AND LIFE

The ocean currents that move water around the planet also carry dissolved minerals that are essential nutrients for the simplest oceanic life forms—tiny plantlike organisms known as phytoplankton. Nearly all other marine life relies on these organisms for food, so in this way the nutrients support the entire oceanic ecosystem. Many of these nutrients are carried into the oceans by rivers. Some of the nutrients are soon taken up by marine life, but released again when the organisms die and their remains settle on the bottom. When ocean floor sediments are stirred up by currents and storms, nutrients are brought back to the surface. Here, the phytoplankton can turn them into food that supports the oceanic food chain.

◀ MAKING FOOD

Microscopic marine bacteria and phytoplankton contain a green substance called chlorophyll that absorbs the energy of sunlight. Using this energy, they combine carbon dioxide with water to make carbohydrates such as sugar and turn these into the tissues that animals use as food. Some deep-water bacteria make food in different ways, but for most oceanic life this process, known as photosynthesis, is the basis of life itself.



◀ VITAL NUTRIENTS

The tiny organisms that make carbohydrates by photosynthesis also need other nutrients. These include the nitrates and phosphates that are essential ingredients of proteins and the oxygen that turns sugar into energy. They need calcium and silica to build their shells and tiny, but essential, amounts of trace elements. Many of these nutrients occur in seawater, but they get incorporated into living things. When these die and their bodies decompose, like the remains of this turtle, the nutrients are released into the water—but many are stored in seafloor sediments until they are stirred up by ocean currents.

nutrients

▶ PLANKTON BLOOMS ▶

Where vital nutrients are available, they nourish the growth of the microscopic plantlike organisms that form the phytoplankton. These tiny flecks of life drift in the sunlit surface waters of all oceans, but they are far more numerous where ocean currents bring abundant nutrients to the surface. In such places, the multiplying phytoplankton form dense plankton blooms that color the water and make it cloudy. They can cover vast areas like this swirling mass of plankton off northern Spain, which is at least 155 miles (250 km) across—and they are usually a sign of rich oceanic life.

